Sandesh Sampath Kumar

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Summary

Backend Engineer experienced in building **high-scale distributed systems** and **AI/LLM-powered backend services** using **Java**, **.NET Core**, **Python**, and **AWS**. Demonstrated impact delivering **8× throughput improvements**, **10× transaction scale**, **$16M+ client revenue**, **95%+ test coverage**, and **55% faster development iteration**. Strong foundation in **microservices**, **cloud-native architectures**, **RAG/LLM systems**, **reliability engineering**, and **observability**.

Technical Skills

Languages:	Java, C#, Python, JavaScript, TypeScript
Backend:	Spring Boot, .NET Core, Node.js, REST APIs, Microservices, Async I/O
AI/ML:	RAG, LLM Integration, Embeddings, Vector Databases (Qdrant), Prompt Engineering
Cloud & Infra:	AWS (Lambda, API Gateway, Step Functions, EC2, RDS, S3), Azure, Docker, Kubernetes, Terraform
Data & Practices:	PostgreSQL, SQL Server, MongoDB, Redis, CI/CD, TDD, Observability
Frontend:	React, Angular, HTML5, CSS3, Bootstrap, Tailwind

Experience

Software Engineer | **Backend & AI** June 2026 – Present
First Help Financial *Boston, MA*

- Own **end-to-end design and delivery** of production backend services, authoring **architecture decision records (ADRs)** to align technical direction across the team.
- Develop and extend core services for the **loan origination system** powering high-volume auto-finance workflows.
- Drive an **AI-first development workflow**, orchestrating **Cursor** with structured **plan/execute prompting** and **memory-file context management**, cutting delivery time per ticket by **35%** while improving code consistency.
- Lead a **RAG-based agentic AI** proof-of-concept to automate **underwriting document review** in the loan origination flow, applying retrieval-augmented pipelines and tool-calling.

Software Engineer Intern | **Backend, Distributed Systems** May 2025 – Dec 2025
First Help Financial *Boston, MA*

- Engineered a reproducible **AWS-local development environment** (SAM + LocalStack), cutting development and debugging time by **55%**.
- Developed **Python libraries** for structured logging, API templates, and automated fault-handling; achieved **95%+ unit test coverage**.

Application Developer | **Backend & Distributed Systems** Sep 2022 – Aug 2024
Thoughtworks *Bengaluru, India*

- Designed distributed **.NET Core** microservices on **Azure** following clean architecture and SOLID principles.
- Scaled an enterprise ordering platform to support **8× higher throughput** and **10× global transaction growth**.
- Launched a developer onboarding portal that cut client integration time by **60%** and enabled **$16M+ in revenue**.
- Boosted system performance by **3×** via profiling, parallel execution, optimized SQL, and refined connection pooling.
- Instrumented services with **Application Insights** for distributed tracing, centralized logging, and real-time dashboards.

Systems Engineer | **Java, Spring Boot, Angular** Aug 2021 – Aug 2022
Tata Consultancy Services *Bengaluru, India*

- Implemented backend functionality for enterprise banking systems using **Spring Boot**, **JPA/Hibernate**, and **SQL**.
- Improved API and database performance by **20%** via targeted query optimization and response-path refinements.
- Built and integrated **Angular** UI components with Spring Boot APIs to deliver end-to-end banking workflows.

Projects

Career Advisor Backend | Python, FastAPI, Qdrant, RAG Sep 2025 – Oct 2025

- Designed a **retrieval-augmented generation** backend using **FastAPI** and **Qdrant** for personalized resume feedback.
- Engineered deterministic pipelines for **chunking**, **embeddings**, **ranking**, and **caching** to ensure reliable LLM-driven responses.

CareerLink Platform | Node.js, React, PostgreSQL Mar 2025 – Apr 2025

- Developed **backend services** for **job matching**, **RBAC**, and **secure authentication** with optimized PostgreSQL filters and indexes.
- Improved query responsiveness through efficient **pagination**, index-driven search paths, and prepared statements.

Education

Northeastern University | **Boston, USA** May 2026
Master of Science in Computer Science *GPA: **4.0/4.0***

Jyothy Institute of Technology | **Bengaluru, India** Aug 2021
Bachelor of Engineering in Computer Science *GPA: **8.66/10***